

```

Last login: Tue Aug 11 12:08:22 on ttys000
minajjareddy@Minajjas-MacBook-Air ~ % nano filecopy.c
minajjareddy@Minajjas-MacBook-Air ~ % clang filecopy.c -o filecopy
minajjareddy@Minajjas-MacBook-Air ~ % echo "Hello, this is a sample file." > sample.txt
minajjareddy@Minajjas-MacBook-Air ~ % ./filecopy sample.txt copy.txt
File copied successfully.
minajjareddy@Minajjas-MacBook-Air ~ % cat copy.txt
Hello, this is a sample file.
minajjareddy@Minajjas-MacBook-Air ~ % ./filecopy sample.txt copy.txt
File copied successfully.
minajjareddy@Minajjas-MacBook-Air ~ % sudo dtruss cat sample.txt
Password:
dtrace: system integrity protection is on, some features will not be available

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dtrace: failed to execute cat: Operation not permitted
minajjareddy@Minajjas-MacBook-Air ~ % >....
    return 1;
}

destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

if (destination == -1) {
    perror("open");
    close(source);
    return 1;
}

while ((bytes_read = read(source, buffer, BUFFER_SIZE)) > 0) {
    bytes_written = write(destination, buffer, bytes_read);

    if (bytes_written != bytes_read) {
        perror("write");
        close(source);
        close(destination);
        return 1;
    }
}

close(source);
close(destination);

printf("File copied successfully.\n");

return 0;
}

```

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zsh: parse error near '\n'
minajjareddy@Minajjas-MacBook-Air ~ % clang filecopy.c -o filecopy
echo "Hello, this is a sample file." > sample.txt
./filecopy sample.txt copy.txt
cat copy.txt
File copied successfully.
Hello, this is a sample file.
minajjareddy@Minajjas-MacBook-Air ~ % USER SPACE
    |
    | open()
    |
    v

```

```

KERNEL SPACE
    |
    | Open file / check permissions
    |
    v
USER SPACE
    |
    | read()
    |
    v
KERNEL SPACE
    |
    | Read file data
    |
    v
USER SPACE
    |

```

```

minajjareddy@Minajjas-MacBook-Air ~ % clang filecopy.c -o filecopy
echo "Hello, this is a sample file." > sample.txt
./filecopy sample.txt copy.txt
cat copy.txt
File copied successfully.
Hello, this is a sample file.
minajjareddy@Minajjas-MacBook-Air ~ % USER SPACE
    |
    | open()
    |
    v

```

```

KERNEL SPACE
    |
    | Open file / check permissions
    |
    v
USER SPACE
    |
    | read()
    |
    v
KERNEL SPACE
    |
    | Read file data
    |
    v
USER SPACE
    |
    | write()
    |
    v
KERNEL SPACE
    |
    | Write file data
    |
    v
USER SPACE
    |
    | close()
    |
    v

```

```

KERNEL SPACE
zsh: parse error near '|'
minajjareddy@Minajjas-MacBook-Air ~ % strace cat sample.txt
zsh: command not found: strace
minajjareddy@Minajjas-MacBook-Air ~ % >....
    |
    |
    v

```

```

    |
    | Open/check file
    |
    v
    |
    | USER SPACE
    |
    | read()
    |
    | v
    |
    | KERNEL SPACE
    |
    | Read data
    |
    | v
    |
    | USER SPACE
    |
    | write()
    |
    | v
    |
    | KERNEL SPACE
    |
    | Write output
    |
    | v
    |
    | USER SPACE
    |
    | close()
    |
    | v
    |
    | KERNEL SPACE
    |
    | Release resource
    |
    v
zsh: parse error near '|'
minajjareddy@Minajjas-MacBook-Air ~ %

```